PO12Q - Quantitative Political Analysis: Uncovering Relationships

Formulae Collection and Notation

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# 1 | Formulae

Table 1: Formulae for PO12Q

| Statistic | Formula |
|---|---|
| **Correlation Coefficients** | |
| Covariance | $\text{cov}(x,y) = \frac{1}{n-1}\sum(x_i - \bar{x})(y_i - \bar{y})$ |
| Pearson's $r$ | $r_p = \frac{\text{cov}(x,y)}{s_x s_y}$ $= \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$ |
| Point-Biserial $r_{pb}$ | $r_{pb} = \frac{\bar{x}_1 - \bar{x}_0}{s}\sqrt{\frac{n_1 n_0}{n(n-1)}}$ |
| Spearman's $r_s$ | $r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$ |
| $\phi$ Coefficient | $\phi = \frac{AD - BC}{\sqrt{(A+B)(C+D)(A+C)(B+D)}}$ |

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| Statistic | Formula |
|---|---|
| **Correlation – Significance Testing** | |
| Noncentrality Parameter (power) | $\lambda = \frac{\rho\sqrt{n-2}}{\sqrt{1-\rho^2}}$ |
| Pearson (Significance) | $t_{\text{obs}} = \frac{r_p\sqrt{n-2}}{\sqrt{1-r_p^2}}$ |
| Point-Biserial (Significance) | $t = \frac{r_{pb}\sqrt{n-2}}{\sqrt{1-r_{pb}^2}}$ |
| Spearman (Significance) | $t = \frac{r_s\sqrt{n-2}}{\sqrt{1-r_s^2}}$ |
| | $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$ |
| $\phi$ (Significance) | $\chi^2 = n\phi^2 \quad (df = 1)$ |
| **Regression** | |
| Adjusted $R^2$ | $\bar{R}^2 = 1 - \frac{\sum \hat{\epsilon}_i^2/(n-k-1)}{\sum(y_i-\bar{y})^2/(n-1)}$ |
| Coefficient of Determination | $R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum \hat{\epsilon}_i^2}{\sum(y_i-\bar{y})^2}$ |
| Estimated Standard Error of $\hat{\beta}_0$ | $\hat{se}(\hat{\beta}_0) = \sqrt{\frac{\sum x_i^2}{n\sum(x_i-\bar{x})^2}}\,\hat{\sigma}$ |
| Estimated Standard Error of $\hat{\beta}_1$ | $\hat{se}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum(x_i-\bar{x})^2}}$ |
| | $\hat{\text{var}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2(\boldsymbol{X'X})^{-1}$ |
| Estimated Variance | $\hat{\sigma}^2 = \frac{\sum \hat{\epsilon}_i^2}{n-p}$ |
| Estimator of $\beta_0$ | $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$ |
| Estimator of $\beta_1$ | $\hat{\beta}_1 = \frac{\sum_{i=1}^N (x_i-\bar{x})(y_i-\bar{y})}{\sum_{i=1}^N (x_i-\bar{x})^2}$ |
| | $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{y})$ |
| Explained Sum of Squares | $\sum(\hat{y}_i-\bar{y})^2$ |
| Residual Sum of Squares | $\sum \hat{\epsilon}_i^2 = \sum(y_i-\hat{y}_i)^2$ |
| Total Sum of Squares | $\sum(y_i-\bar{y})^2$ |
| Variance Inflation Factor | $\text{VIF}_j = \frac{1}{1-\bar{R}_j^2}$ |

## 1.1 Matrices

$$\boldsymbol{X'X} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \end{bmatrix} \times \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}$$

$$(\boldsymbol{X'X})^{-1} = \frac{1}{n\sum x_i^2 - \sum x_i \sum x_i} \begin{bmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{bmatrix}$$

$$\boldsymbol{X'y} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \end{bmatrix} \times \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}$$

# 2 | Notation

Table 2: Notation for PO12Q

| Symbol | Explanation |
|---|---|
| $a$ | A scalar; a single numerical value representing magnitude, written in lower-case italics. |
| $\boldsymbol{a}$ | A vector; an ordered list of numbers (often a column) representing magnitude and direction, written in lower-case bold. |
| $\boldsymbol{A}$ | A matrix; a rectangular array of numbers arranged in rows and columns, written in upper-case bold. |
| $\mathrm{A}^{-1}$ | Inverse of matrix A; the matrix satisfying $AA^{-1} = I$, the matrix analogue of dividing by A. |
| $\mathrm{A}'$ | Transpose of matrix A; the matrix obtained by interchanging its rows and columns. |
| $\alpha$ | Significance level; the probability of rejecting a true null hypothesis (a Type I error). Common levels are 0.05, 0.01, or 0.001; it also fixes the confidence level $1 - \alpha$. |
| $\alpha_i$ | Coefficients of an auxiliary (secondary) regression – one regressing an independent variable on the others, used for example to compute the variance inflation factor (VIF) for multicollinearity rather than to answer the main research question. |
| $\beta$ | Population regression coefficient; the number multiplied by a variable in the regression equation. A slope coefficient gives the change in $y$ for a one-unit increase in that predictor, holding the others constant. |
| $\hat{\beta}$ | Estimated regression coefficient; the Sample Regression Function estimate of $\beta$, obtained by Ordinary Least Squares (OLS), which minimises the sum of squared residuals. |
| $\chi^2$ | Chi-squared test statistic; assesses whether the observed frequencies in a contingency table differ significantly from those expected if there were no association between the variables, $\chi^2 = \sum(O - E)^2 / E$. |
| $d_i^2$ | Difference in ranks for each pair of observations for Spearman Correlation. |
| $d_{\text{obs}}$ | Observed, standardised effect size (Cohen's $d$); the magnitude of the difference, expressed in standard-deviation units. |
| $d_{\text{test}}$ | Assumed effect size, for example in a power or sample-size calculation; the value that enters the noncentrality parameter, $\lambda = d_{\text{test}} \cdot \sqrt{n}$. |
| df | Degrees of freedom, express constraints on our estimation process |
| E[x] | Expected value of $x$; the long-run average of a random variable over infinitely many repetitions, i.e. the mean of its probability distribution. |
| $E[y_i \mid x_i]$ | Conditional expectation, reading as 'The expected value of $y$, given $x$'. |
| $\epsilon$ | Error term; quantifies the distance between each observation and the corresponding point on the (population) regression line, denoted $\epsilon_i$. |
| $\hat{\epsilon}$ | Residual; an estimate of the error term – the difference between an observation $y_i$ and its estimated value $\hat{y}_i$, denoted $\hat{\epsilon}_i$. |
| F | $F$-test statistic; compares explained and unexplained variance to test the joint significance of several coefficients at once. |
| $f^2$ | Cohen's $f^2$; the standardised effect size for an $F$-test, the regression counterpart of Cohen's $d$. For an overall model $f^2 = \frac{R^2}{1-R^2}$, expressing explained relative to unexplained variance; larger $f^2$ means a larger effect and thus greater power. |
| $\frac{\partial f(x)}{\partial x}$ | The partial derivative of $f$ of $x$, with respect to $x$. |
| $\mathrm{H}_0$ | Null Hypothesis, typically asserting that there is no effect or no difference. |
| $\mathrm{H}_1$ | Alternative Hypothesis, proposes that there is an effect or a difference |
| I | Identity matrix; a square matrix with ones on the diagonal and zeros elsewhere, leaving any matrix unchanged under multiplication. |

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Table 2 – continued

| Symbol | Explanation |
|---|---|
| k | Number of slope coefficients (predictors) in the regression model, excluding the intercept. |
| K | Total number of regression coefficients including the intercept, so that $K = k + 1$. |
| $\lambda$ | Noncentrality parameter; how many standard errors the true effect lies from the null value, $\lambda = d_{\text{test}} \cdot \sqrt{n}$; it centres the non-central $t$-distribution under the alternative hypothesis. |
| $\lim_{h \to 0}$ | The limit as $h$ approaches zero. |
| log(x) | Logarithm of the variable $x$; the exponent to which a base $b$ must be raised to yield $x$ (so $x = \log_b n$ when $b^x = n$). In regression the natural logarithm is typically used, for example for skewed variables. |
| $\mu$ | Population mean of the variable of interest; a parameter – a fixed value characterising the population – estimated by the sample mean $\bar{y}$. |
| $\mu_0$ | Null Hypothesis Value |
| $n$ | Sample size; the number of observations in the sample. |
| $n_i$ | Number of cases in each group for Point-Biserial Correlation; $n$ = Total. |
| $\Omega$ | Variance–covariance matrix of the regression errors. Under the standard assumptions $\Omega = \sigma^2 I$, where $\sigma^2$ is the mean squared error. |
| $p$ | p-value, indicates the probability of obtaining a result equal to, or even more extreme than the observed value in the direction of the alternative hypothesis, assuming the null hypothesis is true. |
| $P()$ | Probability of a statement |
| $\phi$ | Phi coefficient; measures the association between two binary variables, equivalent to Pearson's $r$ applied to two dichotomies. |
| $r$ | Correlation coefficient; measures the statistical dependence of two variables by pairwise comparison of values, taking a value between −1 and +1 for strength and direction. |
| $R^2$ | Coefficient of determination; the proportion of variation in the dependent variable explained by the independent variable(s), $R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$. |
| $\bar{R}^2$ | Adjusted $R^2$; a version of $R^2$ that adjusts for the number of independent variables in a multiple regression, giving a more unbiased estimate of the proportion of variance explained. |
| $r_p$ | Pearson's product-moment correlation coefficient; a parametric measure of the linear association between two variables (assumes approximate normality). |
| $r_{pb}$ | Point-biserial correlation coefficient; measures the association between a continuous variable and a binary (dichotomous) variable. |
| $r_s$ | Spearman's rank-correlation coefficient; a non-parametric measure of the monotonic association between two variables, computed on ranks. |
| $\rho$ | The hypothesised population correlation under $H_1$. |
| $s$ | Sample standard deviation; the square root of the variance, $s = \sqrt{\frac{\sum(y_i - \bar{y})^2}{n-1}}$. |
| $se$ | Estimated standard error; the standard deviation of the sampling distribution of $\bar{y}$ estimated from the sample, $se = \frac{s}{\sqrt{n}}$ (replacing the unknown $\sigma$ with $s$). |
| $\hat{se}$ | Estimated standard error; estimates the standard error of a sample regression coefficient. |
| $\sigma$ | Residual Standard Error; measures the average distance that the observed values fall from the regression line, expressed in the same units as the response variable. |
| $\sigma^2$ | Error variance; the variance of the regression disturbances $\epsilon$, assumed constant under homoscedasticity, and estimated by the mean squared error of the residuals. |
| $t_{\text{obs}}$ | Observed, standardised test statistic that measures how many standard errors a sample statistic (typically a sample mean) lies from a hypothesised population parameter. |

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Table 2 – continued

| Symbol | Explanation |
|---|---|
| $u$ | Numerator degrees of freedom of the $F$-distribution; the number of slope coefficients being tested, $u = k$. |
| $v$ | Denominator degrees of freedom of the $F$-distribution; the residual degrees of freedom, $v = n - k - 1$ (equivalently $n - K$). |
| VcV | Variance–covariance matrix; a square matrix holding the variances of the estimated coefficients on its diagonal and the covariances between pairs of coefficients in its off-diagonal elements. |
| $\bar{x}_i$ | Mean of the continuous variable for group $i$ for Point-Biserial Correlation. |
| $\bar{y}$ | Sample mean; the sum of the observations divided by their number, $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$. |
| $\hat{y}$ | A(n estimated) point on the sample regression line. |
| $y_i$ | The $i$-th observation in the sample, where $i = 1, 2, \ldots, n$. |